

A Bayesian Detection of Epistasis
Using Summarized Synergy Networks
in Islet Autoimmunity



Shane E. Ridoux

Advised by Drs. Randi K. Johnson, Joshua P. French & Erin E. Austin

University of Colorado-Denver

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Dedication

To Andrew Gavin Olguin, this is for you.

I could not have done it without you.

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Abstract

Type 1 Diabetes (T1D) is a complex autoimmune disease characterized by the erroneous destruction of islet β -cells resulting in life-long dependence on exogenous insulin. Although T1D is considered to be largely genetic, with the heritability of the disease around 80%, roughly 20% of the phenotypic variance is still unexplained. Epistasis, the interaction of genetic variants, is thought to contribute to this missing heritability in T1D, yet the epistatic landscape of T1D and its precursor Islet Autoimmunity (IA) remains understudied. A genome-wide analysis of the epistatic architecture of IA could inform potential biological pathways in the etiology of T1D in addition to elucidating aspects of its missing heritability.

After quality control, 359 DAISY children selected for nested case-control study with a customized Exome array imputed to the TOPMed r2 reference panel were included for analysis of genome-wide epistasis detection. Within-gene synergy networks were used to capture non-linear epistatic effects and summarized through diffusion kernel principal components (kPC). Between-gene epistasis was detected through a bayesian interaction analysis of kPC-summarized genes.

RESULTS

DISCUSSION

1 Introduction

1.1 Type 1 Diabetes & Islet Autoimmunity

Type 1 Diabetes (T1D) is a complex autoimmune disorder where the immune system erroneously targets and destroys insulin-producing pancreatic islet- β cells, leading to a lack of insulin and elevated blood glucose levels¹⁷. Unlike *monogenic* diseases – like sickle cell disease or cystic fibrosis, which result from variation within a single gene and are relatively rare – *complex* diseases are influenced by multiple chromosomal regions containing genetic variation, known as loci¹. The pathogenesis of T1D is believed to follow a continuous progression, beginning with the appearance of islet autoantibodies, followed by β -cell destruction, the onset of dysglycemia, and ultimately, the manifestation of hyperglycemia-related symptoms. Its preclinical stage, known as Islet Autoimmunity (IA), is characterized by the presence of autoantibodies prior to clinical diagnosis^{17,19}. However, the precise onset of IA remains somewhat ambiguous. The definition used by the Diabetes AutoImmunity Study in the Young (DAISY) cohort classifies IA as the presence of one or more islet autoantibodies on two or more consecutive visits, or a single autoantibody-positive visit followed by a Type 1 Diabetes diagnosis at the next visit¹².

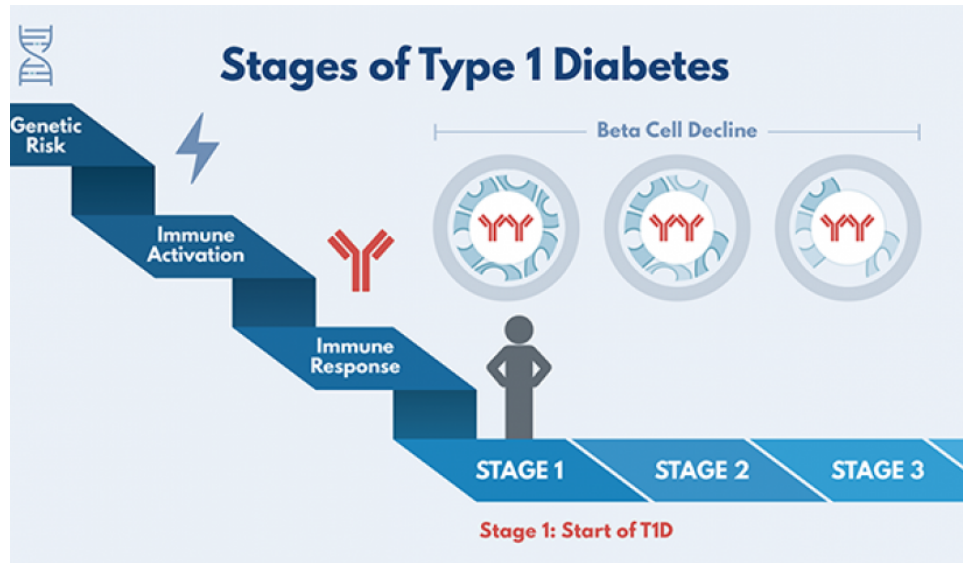


Figure 1: **Stages of T1D**

The progression of T1D follows three recognized stages. **Stage 1** marks the onset of Islet Autoimmunity (IA), with the presence of two or more diabetes-related autoantibodies but normal blood glucose levels. **Stage 2** is characterized by worsening β -cell function and emerging dysglycemia, despite the absence of symptoms. **Stage 3** is the clinical onset of T1D, marked by symptomatic hyperglycemia requiring insulin therapy^{2,19}.

Often diagnosed in early childhood, T1D left untreated can result in serious complications such as diabetic ketoacidosis, a life-threatening condition. With the disease incidence rising⁴, efforts to understand

the pathogenesis and etiology of T1D continue with the hopes of phase-specific therapeutic intervention to mitigate life-threatening complications, eliminate the burden of insulin pumps, and improve quality of life.

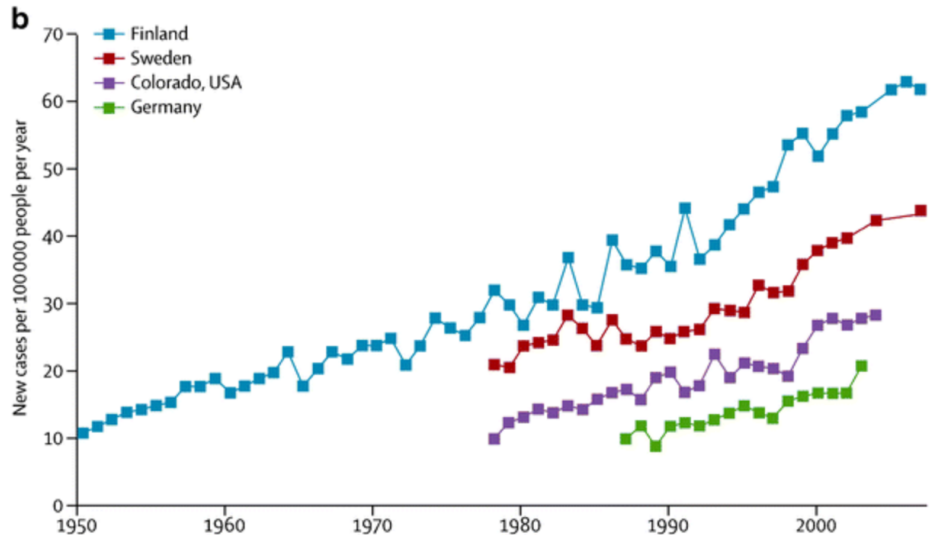


Figure 2: **Incidence of T1D in children aged 0-14 years**

The incidence of T1D has steadily increased across multiple countries over the past several decades, as shown in this longitudinal dataset⁴.

T1D is primarily considered a genetic disease, with environmental factors—such as viral infections—potentially influencing disease progression and the onset of IA. Ongoing research continues to investigate the viral onset hypothesis and the role of the complement system, which is primarily encoded within the Human Leukocyte Antigen (HLA) region¹⁹. However, the focus of this research remains on genetic susceptibility.

The HLA region represents the strongest and longest-known genetic risk factor for T1D. The first association between HLA and juvenile-onset T1D was reported in the early 1970s through serological HLA typing²⁴. This finding was later validated in the 1980s using targeted genetic association testing, which pinpointed specific HLA-DQB1 alleles as key determinants of T1D susceptibility²⁶. This highly polymorphic region has been associated with many other autoimmune diseases and constitutes roughly 40% of the heritability of T1D¹⁹.

Despite decades of research, questions about the genetic architecture of T1D remain. Over 60 candidate loci, collectively explaining roughly 80% of the heritability²⁰, have been identified with respect to T1D. However, this leaves approximately 20% of heritability unexplained, a phenomenon known as the missing heritability problem. Addressing this gap requires further investigation into rare variants, common variants with small effect sizes, and gene-gene interactions (epistasis)^{20,21}.

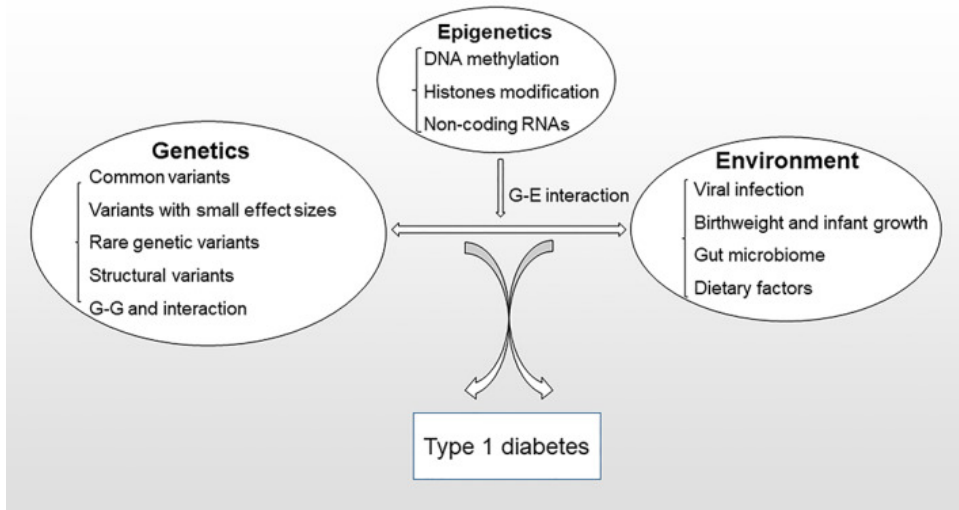


Figure 3: **Potential sources of the missing heritability in T1D**

Here G-G interaction is a gene by gene interaction and G-E is gene by environment interaction²⁰.

1.2 Epistasis

Epistasis has been defined and redefined many times since its coinage by Bateson in 1909, often leading to ambiguity in its definition. Seldom mentioned, however, is that Bateson also defined the term *hypostasis*^{5,8}. These two terms were initially meant to describe a hierarchical relationship where a higher-level factor (epistatic gene/allele) controls whether a lower-level factor (hypostatic gene/allele) can be expressed. Here, *epistatic* and *hypostatic* derive from the Greek prefixes *epi* meaning "upon", *hypo* meaning "under", and root *stasis* meaning "standing". Thus, to Bateson, an epistatic gene "stands upon" a hypostatic gene, masking the hypostatic gene's expression.

Epistasis, or *dual epistacy*, was redefined under a statistical context by Fisher in 1918 as a deviation from linearity in allelic effects with respect to some quantitative phenotype D ^{8,11}. Here, we already see differing definitions of epistasis emerging just 9 years apart, each carrying distinct implications. While Fisherian statistical epistasis does not necessarily imply biological or functional epistasis, it provides a framework for detecting genetic interactions that warrant further investigation. The modern definition of epistasis is largely an umbrella term for genetic interaction. Under this umbrella, we find the term *positive/negative epistasis*, which refers to the interaction between genetic variants where positive (negative) epistasis occurs when their combined effect on a trait or disease risk is greater (less) than the sum of their individual effects. At the population level, this deviation from additivity plays a crucial role in shaping genetic architecture and disease susceptibility¹⁶.

Genome-wide association studies (GWAS) are a modern approach to discover genetic associations with a disease. GWAS test millions of representative single nucleotide polymorphisms (SNPs) from a

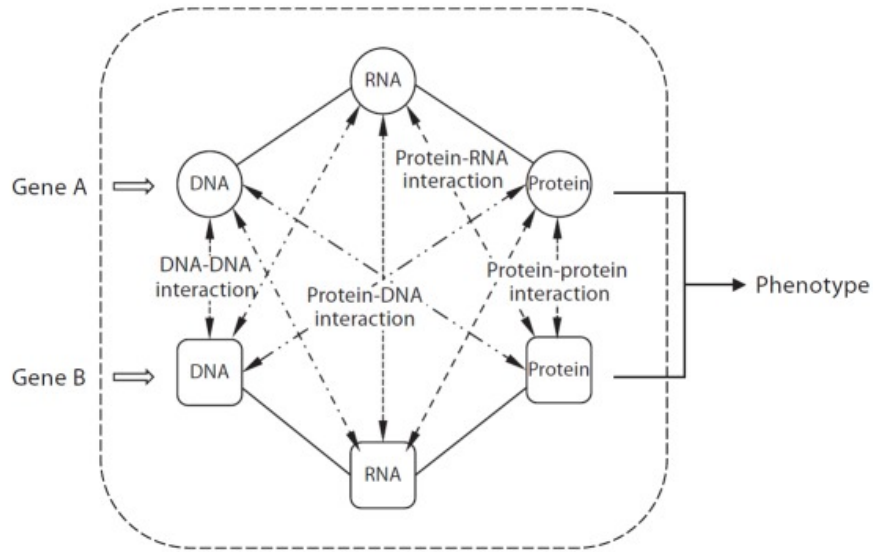


Figure 4: **Gene-Gene Interaction**

This is a conceptual model showing how Gene-Gene interactions play a role in phenotypic variation through multiple molecular interactions, including DNA-DNA, protein-DNA, protein-RNA, and protein-protein interactions²⁷.

population in a linear framework where each SNP is tested for association with a phenotype individually. The standard correction to account for multiple hypothesis testing in GWAS is the Bonferroni correction, which adjusts for the number of independent SNPs in the genome. This method typically applies a genome-wide significance threshold of 5×10^{-8} for European populations to identify statistically significant associations and similar but more stringent threshold for African populations^{9,22}.

There are issues with this standard of practice. The standard genome-wide significance threshold faces challenges in evolving GWAS practices. It was developed for common variants, potentially lacking power for rare variants. It does not take into account the prior probability of a variant being associated with a phenotype, nor does it account for the statistical power of the test which both influence the interpretation of the p-value. These oversights can distort results, particularly when considering epistasis, where interactions between variants may not align with assumptions of independence and testing SNPs in pairs or tuples of length n drastically increase the already large number of hypotheses. Applying the standard threshold universally risks missing meaningful associations while inflating false positives, especially in diverse populations or complex phenotypes^{15,22}.

Genome-wide association interaction (GWAi) studies are an extension of GWAS aimed at epistasis detection^{7,14}. There are two categories of methods that emerge when considering GWAi: exhaustive and

non-exhaustive¹⁴. Non-exhaustive methods avoid the challenge of multiple testing correction by limiting the analysis in some way. Typically, these analyses determine candidate genes a priori to test, use GWAS results to look at interactions between significant SNPs overlooking those SNPs with small effects with potentially large interaction effects, or use some sort of dimensionality reduction^{3,14}. To address the burden of multiple testing in exhaustive approaches, Bayesian methods have been recommended for both GWAS and GWAI^{15,18,22,25,28}.

1.3 Bayesian Statistics

Bayesian statistics is named after Reverend Thomas Bayes, a part-time mathematician who was among the first recorded individuals to attempt solving the problem of inverse probability^{13,23}. Inverse probability seeks to determine the probability of a cause given an observed outcome as opposed to classical probability which estimates the likelihood of an event given known conditions. In Bayes' work that was post-humously communicated by his friend and colleague Richard Price, he posed the problem

"*Given* the number of times in which an unknown event has happened and failed: *Required* the chance that the probability of its happening in a single trial lies somewhere between any two degrees of probability that can be named."⁶

Bayes' original work did not explicitly introduce what we now know as Bayes' theorem:

$$p(\theta|y) = \frac{p(y|\theta)p(\theta)}{p(y)}$$

This was instead formulated by Laplace. Bayes' contribution was rather the concept of compromising prior information with information from observed data. He introduces this concept with an analogy to a billiard table where he considers a scenario where a ball is randomly thrown onto a unit interval, and the goal is to estimate the probability of a future event occurring based on where the ball landed. Bayes introduced what we would now call a uniform prior distribution on θ such that θ was modeled as the probability parameter of a binomial distribution^{6,10}.

**** CONTINUE HISTORY UNTIL idk****

Bayesian approaches incorporate prior probabilities and probabilistic modeling to improve statistical power and control false positives, particularly in the context of epistasis and complex diseases. ****talk about bayesian stats broadly and narrow to our context****

1.4 Motivation

Epistasis is particularly important in complex autoimmune diseases, where interactions between immune-related loci can shape immune system function and the loss of self-tolerance²¹. T1D is a prime example of this. **generally summarize all three background sections to motivate this specific method and how it addresses everything we want it to**

2 Methodology

2.1 Overview

2.2 Data Preparation

Talk about imputation and DAISY Cohort and potentially TEDDY Cohort

2.3 Within-Gene Epistasis Detection

Talk about method generally

2.3.1 Computing Entropy-Based SNP Synergies

The concept of bivariate synergism quantifies the combined effect of two SNPs (A and B) on a disease (D) beyond their individual contributions using information theory. It is calculated as:

$$Syn(A; B; D) = I(A, B; D) - [I(A; D) + I(B; D)]$$

where $Syn(A; B; D)$ compares the joint contribution of SNPs A and B to the disease D with the additive contributions of the individual SNPs. The information gain $I(A; D)$ about the disease D due to knowledge about SNP A and is defined as:

$$I(A; D) = H(D) - H(D|A)$$

$$I(A, B; D) = H(D) - H(D|A, B)$$

where $H(\cdot)$ is the entropy.

$$H(D) = \sum_d p(d) \log\left(\frac{1}{p(d)}\right)$$

$$H(D|A) = \sum_{a,d} p(a, d) \log\left(\frac{1}{p(d|a)}\right)$$

where $p(d)$ is the probability of having disease $D = d$ and $p(d|a)$ is the probability of having disease $D = d$ given SNP A has genotype a . A Network results from the bivariate synergy calculations of each SNP where the edge of nodes A and B has weight $Syn(A; B; D)$.

Entropy can be thought of as a measure for the uncertainty in a

2.3.2 Diffusion Kernels

Why diffusion kernels? Essentially allow for gene summaries that become the explanatory variable in the Bayesian model...

2.4 Between Gene Epistasis Detection

2.4.1 Bayesian Interaction Modeling

Detecting gene-gene interactions

- one continuous variable per gene
- Bayesian semi-parametric regression approach to infer non-linear gene-gene interaction
- As gene explanatory variables for individual i , used first kernel PCA X_{ij_1} for each gene j_1
- genes and gene-gene interactions are selected on Posterior Inclusion Probability (PIP)

Let p be the number of genes, k be a number sufficiently large $\ni$ all exposures are captured in the model, and X_{ij_1} be the first kernel PCA for gene j_1 as gene explanatory variable for individual i .

Assume:

$$Y_i \sim N(f(X_i, \sigma^2)) \quad (1.1)$$

$$f(X_i) = \sum_{h=1}^p f^{(h)}(X_i) \quad (1.2)$$

$$f^{(h)}(X_i) = \sum_{j_1=1}^p \tilde{X}_{ij_1} \beta_{j_1}^{(h)} + \sum_{j_1=2}^p \sum_{j_2 \leq j_1} \tilde{X}_{ij_1 j_2} \beta_{j_1 j_2}^{(h)} \quad (1.3)$$

$$\tilde{X}_{j_1} = g_1(X_{j_1}), \quad \text{where } g_1(\cdot) \text{ is the natural spline basis function} \quad (1.4)$$

3 Results

3.1 Overview

3.2 Plots

3.3 Limitations

4 Discussion

5 Conclusion

6 References

References

- [1] Monogenic Disorder - an overview | ScienceDirect Topics.
- [2] Type 1 diabetes staging classification opens door for intervention | TRIALNET Type 1 Diabetes TrialNet.
- [3] An, P., O. Mukherjee, P. Chanda, L. Yao, C. D. Engelman, C.-H. Huang, T. Zheng, I. P. Kovac, M.-P. Dubé, X. Liang, J. Li, M. de Andrade, R. Culverhouse, D. Malzahn, A. K. Manning, G. M. Clarke, J. Jung, and M. A. Province (2009). The Challenge of Detecting Epistasis ($G \times G$ Interactions): Genetic Analysis Workshop 16. *Genetic epidemiology* 33(0 1), S58–S67.
- [4] Atkinson, M. A., G. S. Eisenbarth, and A. W. Michels (2014, January). Type 1 diabetes. *Lancet (London, England)* 383(9911), 69–82.
- [5] Bateson, W. (1909). *Mendel’s Principles of Heredity*. Cambridge, Massachusetts London, England: Cambridge University Press.
- [6] Bayes, T. and R. Price (1997, January). LII. An essay towards solving a problem in the doctrine of chances. By the late Rev. Mr. Bayes, F. R. S. communicated by Mr. Price, in a letter to John Canton, A. M. F. R. S. *Philosophical Transactions of the Royal Society of London* 53, 370–418. Publisher: Royal Society.
- [7] Bessonov, K., E. S. Gusareva, and K. Van Steen (2015, July). A cautionary note on the impact of protocol changes for genome-wide association SNP $\times$ SNP interaction studies: an example on ankylosing spondylitis. *Human Genetics* 134(7), 761–773.
- [8] Cordell, H. J. (2002, October). Epistasis: what it means, what it doesn’t mean, and statistical methods to detect it in humans. *Human Molecular Genetics* 11(20), 2463–2468.
- [9] Evangelou, E. (Ed.) (2018). *Genetic Epidemiology: Methods and Protocols*, Volume 1793 of *Methods in Molecular Biology*. New York, NY: Springer New York.
- [10] Fienberg, S. E. (2006, March). When did Bayesian inference become "Bayesian"? *Bayesian Analysis* 1(1), 1–40. Publisher: International Society for Bayesian Analysis.
- [11] Fisher, R. A. (1919). XV.—The Correlation between Relatives on the Supposition of Mendelian Inheritance. *Transactions of the Royal Society of Edinburgh* 52(2), 399–433.

- [12] Frohnert, B. I., L. Ide, F. Dong, A. Barón, A. K. Steck, J. Norris, and M. J. Rewers (2017, June). Late-onset islet autoimmunity in childhood: the Diabetes Autoimmunity Study in the Young (DAISY). *Diabetologia* 60(6), 998–1006.
- [13] Gelman, A., J. B. Carlin, H. S. Stern, D. B. Dunson, A. Vehtari, and D. B. Rubin. Bayesian Data Analysis Third edition (with errors fixed as of 15 February 2021).
- [14] Gusareva, E. S. and K. Van Steen (2014, November). Practical aspects of genome-wide association interaction analysis. *Human Genetics* 133(11), 1343–1358.
- [15] Jiang, X., M. M. Barmada, G. F. Cooper, and M. J. Becich (2011). A bayesian method for evaluating and discovering disease loci associations. *PloS One* 6(8), e22075.
- [16] Johnson, M. S., G. Reddy, and M. M. Desai (2023, May). Epistasis and evolution: recent advances and an outlook for prediction. *BMC Biology* 21(1), 120.
- [17] Katsarou, A., S. Gudbjörnsdottir, A. Rawshani, D. Dabelea, E. Bonifacio, B. J. Anderson, L. M. Jacobsen, D. A. Schatz, and Å. Lernmark (2017, March). Type 1 diabetes mellitus. *Nature Reviews Disease Primers* 3(1), 1–17. Publisher: Nature Publishing Group.
- [18] López, M. J. M. Bayesian Statistical Methods for the Detection of Epistasis.
- [19] Onengut-Gumuscu, S., B.-J. M. Webb-Robertson, S. Sarkar, A. Manichaikul, X. Hu, A. Frazer-Abel, V. M. Holers, M. J. Rewers, and S. S. Rich (2024). Genetic variants in the complement system and their potential link in the aetiology of type 1 diabetes. *Diabetes/Metabolism Research and Reviews* 40(1), e3716. _eprint: <https://onlinelibrary.wiley.com/doi/pdf/10.1002/dmrr.3716>.
- [20] Pang, H., J. Lin, S. Luo, G. Huang, X. Li, Z. Xie, and Z. Zhou (2022, October). The missing heritability in type 1 diabetes. *Diabetes, Obesity & Metabolism* 24(10), 1901–1911.
- [21] Rose, A. M. and L. C. K. Bell (2012, October). Epistasis and immunity: the role of genetic interactions in autoimmune diseases. *Immunology* 137(2), 131–138.
- [22] Sham, P. C. and S. M. Purcell (2014, May). Statistical power and significance testing in large-scale genetic studies. *Nature Reviews Genetics* 15(5), 335–346. Publisher: Nature Publishing Group.
- [23] Shoemaker, J. S., I. S. Painter, and B. S. Weir (1999, September). Bayesian statistics in genetics: a guide for the uninitiated. *Trends in Genetics* 15(9), 354–358.
- [24] Singal, D. P. and M. A. Blajchman (1973, June). Histocompatibility (HL-A) Antigens, Lymphocytotoxic Antibodies and Tissue Antibodies in Patients with Diabetes Mellitus. *Diabetes* 22(6), 429–432.

- [25] Stephens, M. and D. J. Balding (2009, October). Bayesian statistical methods for genetic association studies. *Nature Reviews Genetics* 10(10), 681–690.
- [26] Todd, J. A., J. I. Bell, and H. O. McDevitt (1987, October). HLA-DQB gene contributes to susceptibility and resistance to insulin-dependent diabetes mellitus. *Nature* 329(6140), 599–604. Publisher: Nature Publishing Group.
- [27] Wang, X., R. C. Elston, and X. Zhu (2010). The meaning of interaction. *Human Heredity* 70(4), 269–277.
- [28] Zhou, X., P. Carbonetto, and M. Stephens (2013, February). Polygenic Modeling with Bayesian Sparse Linear Mixed Models. *PLoS Genetics* 9(2), e1003264.

A Data Preprocessing

B Code

C Additional Figures & Tables